FELIPE RIBEIRO

feliperi@buffalo.edu | 718-603-1100 | linkedin.com/in/fevillasboas | Buffalo, NY | Queens, NY

EDUCATION

University at Buffalo, State University of New York — **Buffalo, NY**
Bachelor of Science in Mechanical Engineering, Minoring in Robotics — *Expected May 2027*
Relevant Coursework: Finite Element Analysis, Control Systems, Thermodynamics, Heat Transfer

TECHNICAL SKILLS

Robotics & Embedded: ESP32, Raspberry Pi, PID Tuning, Motor & energy modeling
Programming & Tools: MATLAB, Coppelia Sim, Arduino IDE, C++ MS Office (Excel, Word, PowerPoint)
CAD & Analysis: Fusion 360, SolidWorks, FEA

ENGINEERING EXPERIENCE

UB ASME — SDC Team ("PODIUM" Design Competition) — **Buffalo, NY**
Co-Lead, Payload Electronics & Automation Division — *June 2026 – Present*

- Co-leading UB ASME's first-year entry into the "PODIUM" Student Design Competition, directing a 12-person Payload Electronics & Automation division covering drivetrain, lift, launcher, and autonomy.
- Designing the robot's four-rail power system (motor bus, 6V servo, 5V logic, 3.3V logic) off a 4S battery with a ground-star layout, and documenting pin-level wiring for the drive, lift, and sensing subsystems.
- Selecting autonomy sensors (IR line array, ultrasonic and time-of-flight ranging, IMU) and ESP32 control for the 30-second autonomous phase.
- Compared pneumatic and flywheel launchers; wrote the budget request that raised the team's budget from $500 to $750.

UB ASME — IAM3D 2026 Rover Team — **Buffalo, NY**
Electronics Lead — *August 2025 – Present*

- Leading electronics integration for a 3D printed rover competition, coordinating across the mechanical, electrical, and software sub-teams; evaluated on payload delivery effectiveness against competing colleges.
- Designed, wired, and documented (schematics + test procedures) a multi-battery power system, increasing rover motor performance by ~25% while regulating voltage for 6.25–8.25V high-torque servos.
- Programmed Arduino and ESP32 to translate RC receiver signals to ESC-driven DC motors, servos, relays, and onboard subsystems.
- Built and integrated a servo-actuated excavator scoop system with FPV camera support, laser alignment assistance, and custom 3D-printed protective enclosures.
- Competed with the IAM3D rover team at the EFX engineering competition (University of Central Florida).

TECHNICAL PROJECTS

Cycle Ball Intake & Flywheel Launcher System — **Buffalo, NY**
UB ASME SDC Team — Payload Subsystem — *October 2026 – Present*

- Designed a 25 mm × 150 mm belt-driven intake roller in Fusion 360 with six ball-guiding fins and 16-tooth 3D-printed pulleys that feeds 65 mm balls into a gravity-fed chamber with a notch that holds them in place.
- Designed the release so removing the notch lets balls roll into a flywheel shooter that fires over the robot; the number of shots is set by how long the notch stays open.
- Modeling motor choice, battery use, flywheel inertia, and belt sizing in MATLAB/Python for the feed and launch rate.

Forklift Lift & Telescoping Mast — **Buffalo, NY**
UB ASME SDC Team — Payload Subsystem — *June 2026 – Present*

- Designing a forklift lift on aluminum extrusion with dual lead-screw Z axes driven by a stepper, like an open-frame 3D printer Z stage, with limit switches for homing and tier stops.
- Designing a cable-driven telescoping mast to extend the lift's reach to the higher dropoff tiers.

Four-Bar Linkage Prosthetic Hand — **Buffalo, NY**
Personal Project — *June 2026 – Present*

- Designing a prosthetic hand in which each finger is driven by a four-bar linkage that converts a single 180° servo's rotation into controlled finger curl via a fishing-line tendon for inward and outward articulation.
- Modeled and validated the finger linkage kinematics in PMKS+ to tune range of motion and mechanical advantage before physical prototyping.
- Currently designing and 3D-printing individual finger assemblies, integrating servo actuation and tendon routing into the hand structure.

Self-Balancing Inverted Pendulum Robot — **Buffalo, NY**
Personal Project — *February 2026*

- Built a 3D-printed two-wheel inverted pendulum robot using (Arduino, MPU-6050, DRV8833), cutting balance sway from +8°/−6° to a steady ±4.5° by adding mass to raise inertia and rigidly mounting the IMU to reduce tilt-estimate noise.
- Fused accelerometer and gyroscope data via a two-state Kalman filter feeding a PID loop (Kp 90, Ki 0.2, Kd 6) for real-time balance and disturbance rejection.
- Through structured experimental testing and data analysis, diagnosed mismatched startup thresholds between the two TT gearmotors; measured each motor's activation point and used the map() function to scale PID output across both motors' usable PWM range, giving the controller finer resolution and better wheel response.

PROFESSIONAL EXPERIENCE

Infusion by Castles — **New York, NY**
Advertising Intern — Client Operations & Office Design — *August 2022 – September 2024*

- Organized direct mail campaign files by market segment (International, AA, HS) for client access, and led an office layout redesign during a company relocation to support workflow efficiency.